

DENGQEN, China

WMO: 561160

Lat: 31.4147N

Lon: 95.5936E

Elev: 3874

StdP: 62.66

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-13.0	-27.4	0.5	-3.3	-25.3	0.6	-4.1	9.6	-0.7	8.2	-0.6	0.4	290	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	21.9	10.3	20.7	9.9	19.4	9.6	11.8	18.4	11.2	17.7	10.7	16.8	2.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
9.7	12.2	13.5	9.1	11.7	13.0	8.6	11.3	12.7	47.1	18.5	45.4	17.9	43.7	17.0	14.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.2	6.4	5.2	-18.3	24.7	1.6	0.9	-19.5	25.3	-20.5	25.8	-21.4	26.3	-22.6	26.9	DB
			-19.4	12.9	1.5	0.7	-20.4	13.4	-21.3	13.8	-22.1	14.1	-23.2	14.7	WB

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-4.9	-2.7	0.4	4.4	8.5	11.9	13.4	13.1	10.7	5.2	-0.1	-3.8
	DBStd	6.99	2.87	2.66	2.69	2.31	2.55	2.43	2.00	2.04	2.14	3.04	2.56	2.63
	HDD10.0	2257	462	355	297	167	61	10	2	2	17	151	304	428
	HDD18.3	4970	720	588	556	417	305	194	152	161	229	407	554	687
	CDD10.0	329	0	0	0	0	14	66	108	100	38	2	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	6	0	0	0	0	0	1	2	3	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	1.4	1.4	1.6	1.7	1.7	1.7	1.5	1.3	1.2	1.2	1.2	1.2	1.2
Precipitation	PrecAvg	636	3	8	13	24	53	128	140	121	99	37	8	3
	PrecMax	885	11	24	37	68	112	217	209	257	162	86	29	15
	PrecMin	378	0	0	1	3	8	51	47	24	37	7	0	0
	PrecStd	122	3	6	9	15	26	44	44	63	33	22	8	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	9.6	13.8	16.3	20.2	22.7	23.4	23.5	21.3	18.1	11.7	9.3
		MCWB	-1.3	-1.3	1.3	4.4	8.0	10.0	11.0	10.1	9.4	6.9	0.8	-1.1
	2%	DB	6.2	7.4	10.2	13.9	18.1	20.8	21.8	21.9	19.2	15.2	9.4	6.4
		MCWB	-2.6	-1.8	0.5	3.1	6.7	9.6	10.7	10.1	9.0	5.8	0.1	-2.2
	5%	DB	3.6	5.5	8.3	12.1	16.4	19.2	20.4	20.5	17.7	13.2	7.7	4.4
		MCWB	-3.8	-2.6	-0.4	2.2	5.8	9.1	10.4	10.0	8.6	4.9	-0.6	-3.0
	10%	DB	1.5	3.7	6.6	10.4	14.6	17.6	18.9	18.8	16.2	11.2	6.0	2.6
		MCWB	-4.8	-3.4	-1.1	1.7	5.1	8.9	10.2	9.7	8.4	4.1	-1.2	-4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.5	-0.1	2.6	5.7	9.3	11.8	12.7	12.3	11.0	8.0	2.1	-0.5
		MCDB	8.0	7.6	10.8	14.4	17.7	18.6	19.8	19.1	17.9	15.0	8.3	7.4
	2%	WB	-2.3	-1.3	1.3	4.2	8.0	11.0	12.0	11.6	10.2	6.9	0.9	-1.8
		MCDB	5.1	5.7	8.7	11.7	15.6	18.0	18.8	18.2	16.6	13.7	7.4	5.6
	5%	WB	-3.5	-2.2	0.2	3.2	7.0	10.4	11.5	11.1	9.6	5.9	0.0	-2.8
		MCDB	3.0	4.4	6.6	9.9	14.1	16.9	17.9	17.4	15.6	11.4	6.2	3.6
	10%	WB	-4.6	-3.1	-0.6	2.4	6.1	9.7	11.0	10.5	9.0	4.9	-0.8	-3.7
		MCDB	0.9	2.9	5.1	8.6	12.5	15.7	17.0	16.5	14.6	9.5	4.9	2.0
Mean Daily Temperature Range	5% DB	MDBR	13.1	12.0	11.1	11.2	10.9	10.3	10.1	10.5	10.8	10.7	12.3	13.7
		MCDBR	17.1	15.6	14.1	14.2	13.9	13.3	13.3	14.0	13.5	13.7	15.0	16.5
		MCWBR	10.3	8.7	6.8	6.0	5.0	4.5	4.2	4.4	4.9	5.3	7.3	9.5
	5% WB	MCDBR	15.8	13.9	12.3	11.9	11.8	10.5	10.3	10.3	10.7	11.8	13.3	15.5
		MCWBR	9.7	8.1	6.2	5.3	4.7	4.2	3.8	3.8	4.4	4.9	6.8	9.2
Clear-Sky Solar Irradiance	taub	0.215	0.245	0.301	0.322	0.308	0.289	0.287	0.285	0.266	0.244	0.213	0.198	
	taud	2.558	2.460	2.286	2.281	2.422	2.589	2.645	2.630	2.661	2.641	2.634	2.656	
	Ebn at Noon	1046	1030	982	967	977	989	991	994	1011	1022	1044	1060	
	Edh at Noon	80	99	127	134	117	99	93	93	86	81	74	69	
All-Sky Solar Radiation	RadAvg	3.63	4.42	5.33	5.91	6.29	5.77	5.66	5.34	4.87	4.19	3.99	3.51	
	RadStd	0.27	0.19	0.24	0.32	0.35	0.35	0.35	0.48	0.32	0.33	0.20	0.23	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	+0.31	N/A	-1.98	N/A	-0.29	N/A	-118	-164	N/A
(47)	Regional (3 neighbors)	+0.42	+0.61	-1.79	N/A	-0.23	N/A	N/A	-156	+80

Nomenclature: See separate page